Hulki Çıray

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Education

Koç University Istanbul, Turkey
Master of Science in Computer Science and Engineering | CGPA: 3.29 / 4.00 *2023 – 2026*
- **Thesis:** SWIFT: Selective Weight Identification and Fine-Tuning | (Link)
- **Advisor:** Prof. Gözde Gül Şahin
- **Research Affiliation:** KUIS AI Center Graduate Research Fellow

Istanbul Technical University (Ranked 91st Globally, QS World University Rankings 2026) Istanbul, Turkey
Bachelor of Science in Electrical Engineering | CGPA: 3.02 / 4.00 *2016 – 2021*
- **Graduation Project:** Object Detection with Deep Learning for Autonomous Cars
- **Advisor:** Prof. Tahir Çetin Akıncı

Izmir Ataturk High School (Ranked 7th Nationwide) Izmir, Turkey
Math-Science Track *2011 – 2015*

Research Interests

Efficient & Reliable Adaptation: LLM Post-Training, Selective Parameter Fine-Tuning, Model Merging, Continual Adaptation without Forgetting, Efficient Model Training
Multimodal & Embodied AI: Vision-Language Models, Vision-Language-Action Models, Language&Vision for Robotics
Trustworthy & Distributed AI: Federated Learning, Privacy-Preserving Domain Adaptation, Explainable AI

Publications & Manuscripts

Hulki Çıray, Gözde Gül Şahin. *SWIFT: Selective Weight Identification and Fine-Tuning.* In preparation for ICML 2027.
Hulki Çıray. *A Comparative Analysis of U-Net Variants for Image Segmentation.* Technical Report, KUIS AI Center. (Link)

Research Experience

Visiting Researcher Sep 2026 – Present
ILS Lab, FAU – Advisor: Prof. Gözde Gül Şahin *Erlangen-Nürnberg, Germany*
- Leading the extension of **SWIFT** into a full conference paper for submission to ICML 2027.
- Running large-scale LLM fine-tuning and model merging experiments on the MareNostrum 5 supercomputer (Barcelona Supercomputing Center).

Graduate Research Assistant Feb 2025 – Aug 2026
TÜBİTAK Project (Grant No. 121C132) – Koç University *Istanbul, Turkey*
- Developed **SWIFT**, a selective parameter-efficient fine-tuning framework that identifies and updates only a small subset of weights to eliminate parameter redundancy and mitigate catastrophic forgetting during domain adaptation.
- Designed a merging mechanism as an extension to SWIFT that prevents parameter interference when combining adapted models.
- Investigated privacy-preserving language adaptation of multilingual LLMs in edge-device federated learning settings using PyTorch and Flower.
- Designed and evaluated model compression techniques (quantization, distillation, pruning) and Retrieval-Augmented Generation (RAG) methods.
- Worked on the reliability of trustworthy agentic AI systems.

Graduate Research Fellow Oct 2023 – Aug 2026
KUIS AI Center *Istanbul, Turkey*
- Collaborated on foundational NLP research initiatives, conducting experiments in both NLP and computer vision.
- Investigated long-context modeling strategies and state-space architectures for large language models.
- Conducted instruction-tuning experiments and inference optimizations using vLLM for domain-specific NLP benchmarks.
- Conducted a comparative analysis of U-Net variants for image segmentation, systematically evaluating accuracy–efficiency trade-offs to identify optimal model configurations.

Computer Vision Research Engineer Aug 2021 – Sep 2021
Baykar Defense, Computer Vision R&D Department *Istanbul, Turkey*
- Implemented real-time deep visual tracking and object detection models using PyTorch for autonomous unmanned aerial vehicles (UAVs).

Teaching Experience

Lead Teaching Assistant Jan 2026 – Jun 2026
Modern Natural Language Processing – FAU Erlangen-Nürnberg *Erlangen, Germany*
- Assisted in course delivery, lab preparation, student project mentoring, and live PyTorch coding sessions on advanced NLP topics.

Teaching Assistant Jan 2026 – Jun 2026
COMP 125: Introduction to Programming with Python – Koç University *Istanbul, Turkey*
- Led weekly live coding sessions covering fundamental Python programming and data structures.

Industry Experience

Senior AI Consultant May 2025 – Dec 2025
Badael Company, Data & AI Team *Riyadh, Saudi Arabia*
- Coordinated cross-functional vendor teams (14 engineers) on production ML system architecture and deployment pipelines.
- Provided technical advisory on open-source LLM selection, PEFT strategies, and inference optimization for enterprise AI deployment.
- Advised the TAZI AI team led by Prof. Zehra Çataltepe on open-source LLM selection, PEFT strategies, and inference optimization for an internal chatbot deployment.
- Directed external engineering teams and contributed to the design and deployment of multi-agent systems using LangGraph.

Machine Learning Engineer Nov 2021 – Jan 2025
Hepsiburada, R&D Department *Istanbul, Turkey*
- Developed scalable machine learning pipelines for customer segmentation, churn prediction, and LTV estimation for over 70 million customers.
- Built a discount sensitivity model using LightGBM and deployed automated feature and inference pipelines via Apache Airflow, improving conversion rates by 1.4% while maintaining gross margin targets and reducing manual discount calibration time by 95%.
- Implemented a Two-Tower deep neural network architecture for personalized product recommendations.

Honors, Grants & Fellowships

TÜBİTAK 2211/2210-A National Graduate Fellowship, Scientific and Technological Research Council of Turkey
TÜBİTAK Project Research Grant (Project ID: 121C132), Graduate Research Funding, Koç University
Merit-Based Scholarship, 100% Tuition Waiver and Monthly Stipend, Koç University
Graduate AI Fellowship, KUIS AI Center, Koç University
Dean's High Honor List, Koç University (Spring 2025)
Dean's Honor List, Koç University (Fall 2023, Spring 2024)
Dean's High Honor List, Istanbul Technical University (Spring 2020)
Dean's Honor List, Istanbul Technical University (Fall 2018, Spring 2019, Fall 2020)

Skills, Languages & Test Scores

Programming & Frameworks: Python, PyTorch, HuggingFace Transformers, vLLM, LangGraph, Flower (Federated Learning), TensorFlow, Scikit-learn, C/C++ (basic), CUDA (basic), Vector Databases, SQL, GCP, Git, Apache Airflow
High-Performance Computing: GPU training of LLMs on the MareNostrum 5 supercomputer (Barcelona Supercomputing Center) and the Koç University HPC cluster, SLURM
Languages: Turkish (Native), English (Advanced)
Standardized Test Scores: GRE Quantitative: 164/170